

**AGREEMENT FOR INDEPENDENT SYSTEM AND MARKET
OPERATOR (ISMO) WEBSITE DEVELOPMENT**

**Agreement for the ISMO website development
including Documentation, Training, three years
Domain registration & Hosting and Initially one
year Support**

**Between
Central Power Purchasing Agency Guarantee Limited
(CPPA-G)**

&

M/s Nextbridge pvt Ltd.

Dated: 15-December-2021

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Description : AGREEMENT OR MEMORANDUM OF AN AGREEMENT - 5(ccc)
First Party : Nextbridge Pvt Ltd [30000-0000000-0]
Second Party : CPPA G [30000-0000000-0]
Agent : Muhammad Uzair Faiz [35101-1669625-3]
Stamp Duty Paid by : CPPA G [30000-0000000-0]
Issue Date : 15-Dec-2021, 12:14:36 PM
Paid Through Challan : 202167F1EBA12D9C
Amount in Words : Five Hundred Rupees Only

Please Write Below This Line

THE PARTIES

Contractor = M/S NEXTBRIDGE PVT LTD.
Central Power Purchasing Agency (Guarantee) Limited = (the “CPPA-G”)

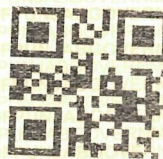
1. BACKGROUND

The Central Power Purchasing Agency (Guarantee) Limited (hereinafter referred to as the “CPPA-G”) invited sealed quotations under the Single Stage – Two Envelope method (i.e., with technical and financial bids submitted separately) from the potential bidders for the ISMO website development including Documentation, Training, Three years Domain registration & Hosting and Initially one year Support. Pursuant to the tender No. CPPA-G/2295-99, dated 27th May 2021, wherein the tender was won by M/S NEXTBRIDGE PVT LTD. (hereinafter referred to as the “Contractor”).

2. SCOPE OF WORK

- Because of the successful Contractor of the tender, this Agreement is executed *inter se* the Parties for the ISMO website development including Documentation, Training, three years Domain registration & Hosting and Initially one year Support of the CPPA-G, Islamabad by the Contractor. Project scope is to build ISMO online presence by developing web application for public and Registration Parties. This application would provide easy access to ISMO staff to share and post information. Through this application Registered Parties would be able to review their billing statements.

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E-STAMP
CONTINUATION SHEET

For the full understanding and realization of the Scope of Work (detailed in Annexure: I). Products / services which are not specifically requested in this Agreement or the annexures there to, but which are necessary to provide for the efficacious functional capabilities proposed by the Contractor through its bid shall also be included therein and shall be read into this Agreement by the Parties.

- ii. **Development Phases:** ISMO online web application will be developed in 2 phases. In Phase 1 all feature set related to CMS website will be completed. The Contractor will be responsible to includes Front-end application for public and admin panel for ISMO staff. Landing page will be kept simple and sophisticated with images been localized. Look and feel of website will be clean, sleek, and minimalistic. A dynamic menu structure will be provided that allows users to effortlessly add, update, move, and delete menu items across the site's structure. The Contractor will provide ISMO Library for documents related to regulations, announcements, reports etc. A dynamic menu structure will be provided that allows users to effortlessly add, update, move, and delete menu items across the site's structure. An activity-based logs will be maintained for all roles. Automatic sitemap will be updated whenever the content creator adds, edits, or deletes any content. From admin panel, admin users will have option (checkbox) to convert non-PDF documents into PDF while uploading document. The Contractor will be responsible to develop a Complaint management tool that will be available for Registered Parties to generate complaints and ISMO department to review and manage these complaints. From dashboard, RP would be able to view Total no. of complaints, Total no. of complaints resolved, Total no. of complaints unresolved. Along with this, the contractor will provide a simple contact us form will be provided where user will enter his personal details along with query. Registered Parties will have the option to receive important notices through email. There will be a FAQ section on admin portal. Job posting section will be available on ISMO for recruitment process. ISMO website would be multi-lingual website. Primary language would be English. Secondary language would be Urdu. 1, ISMO online website will have basic online registration FORM where Registered Parties would be able to register and provide company information. Along with providing information required for registration, Registered Parties can upload documents. Social Media accounts Facebook, Twitter and LinkedIn will be integrated with the website. Functionality of ChatBot will be implemented where user will be able to ask query and ChatBot will answer accordingly to available knowledgebase. Guest user will be able to subscribe for RSS feed. In Phase 2, the contractor will be responsible to develop Registered Parties portal along with 3rd party API integrations. Complaint Management and Discussion Forum features will be included in phase 2 development. Layout of front-end will be developed in phase 1.
- iii. The overall website will be responsive and will support Apple Safari, Google Chrome, Microsoft Internet Explorer (version 11) and Edge, and Mozilla Firefox. DDoS and other cyberattacks will be mitigated and a mechanism will be deployed to detect malicious activities. Code and database backup will be maintained. SSL certificate will be installed on server to encrypt data in site communications.
- iv. The Contractor is responsible for ensuring the completion of all the tasks delineated in the Scope of Work (hereinafter, the "Project").

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3. TIME SCHEDULE

4.1) Effective Date of Commencement

Effective date of Commencement of Service under this Agreement is 01-02-2022 (01st February 2022).

4.2) Time Schedule of Services

The time schedule of Services is given in the attached Annexure: III.

4.3) Extension of Time

Extension of Time for completion of Services and the terms & conditions thereof shall be mutually agreed between the Client and the Contractor as and when required.

4. PROJECT PLAN

The project plan for the required services is given in the attached Annexure: II.

5. PAYMENTS & TAXES

- i. The Contractor shall be entitled to payments by CPPA for the fulfillment of the Scope of Work, in accordance with Annexure: IV. It is agreed and understood by the Contractor that liquidated damages shall be applicable for any delays in the fulfillment of the Scope of Work, and the Parties agree that any liquidated damages imposed under this Agreement are not a penalty but are a genuine pre-estimate of any losses incurred because of such delay(s).
- ii. The quoted unit prices and the total price (inclusive of all taxes and duties where applicable) shall be as per the price schedule in accordance with the bidding document(s), with delivery effected at the designated location(s) at CPPA-G, Islamabad specified in the purchase order. CPPA-G reserves the right to increase / decrease the Scope of Work by up to fifteen percent (15%) at the time of the award / during the currency of this Agreement. No request for an increase in price(s) due to market fluctuation in the cost of goods and services shall be entertained.
- iii. If there is no mention of taxes, the offered / quoted price shall be considered inclusive of all prevailing and applicable taxes / duties. The Contractor shall be responsible for raising an invoice(s) separately for all new taxes, if any, levied by the Government until the completion of term of this Agreement. Any benefit of exemption from or reduction in any prevailing and applicable taxes and duties shall be passed on to CPPA-G.

6. PERFORMANCE CERTIFICATE

The performance certificate shall be issued by Chief Information Officer CPPA-G or his duly authorized representative after the satisfactory completion of each year of the support. The Contractor shall apply for the Performance Certificate at least thirty (30) days prior to the planned completion of the given year of the Project and shall submit a report of the progress to date specifically detailing the progress made in the given year of the support.

7. PERFORMANCE SECURITY AND FORFEITURE THEREOF

- i. The Contractor submitted a Performance Security valid for a period until the completion of Contract upon the receipt of the last consignment, equal to five percent (5%) of the total value (inclusive of GST) of the Project. For avoidance of doubt, the Performance Security shall only be returned after the successful completion of the Project.
- ii. CPPA-G shall have the right to call upon the Performance Security, and the Contractor agrees to forfeit the Performance Security:
 - if the Contractor fails to services specified in the Scope of Work within the time specified
 - if the Contractor commits any breach of this Agreement
 - For other reasons for forfeiting the Performance Security specified in the award letter issued by CPPA-G.

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- iii. If the forfeiture of the security deposit does not adequately compensate CPPA-G for any loss(es) incurred due to the non-delivery of goods / services or breach of this Agreement or for any other reasons including unfair practice(s) or misrepresentation, CPPA-G reserves the right to avail any or all remedies or actions available to it under the law.

8. RESOLUTION OF DISPUTES

In case of any dispute concerning the interpretation and / or application of this Agreement, the matter shall be settled through arbitration in accordance with the provisions of the Arbitration Act, 1940. The seat and venue of such arbitration shall be Islamabad, Pakistan.

9. CORRUPTION AND FRAUD

Indulgence in corrupt and fraudulent practices is liable to result in the termination of this Agreement, the calling upon of the Performance Security by CPPA-G, and the debarring and blacklisting of the Contractor for a stated or indefinite period. CPPA-G also reserves the right to avail any or all remedies or actions available to it under the law.

10. VALIDITY & RENEWAL

The Agreement is intended till the completion of Project.

- i. The duration of the Support Period will initially be one (1) Agreement year, extendable for a period of one (1) Agreement year per extension, for a maximum of three (3) Agreement years. Support Period will start after issuance of Final Acceptance Letter.
- ii. Subject to clause (i) of The Agreement, if M/s Nextbridge Pvt Ltd. can achieve a Satisfactory Annual Performance Certificate from the Chief Information Officer of CPPA-G (the "CIO, CPPA-G") or his duly authorized representative at the end of each Agreement year, the Agreement shall automatically be extended for a period of one (1) Agreement year.

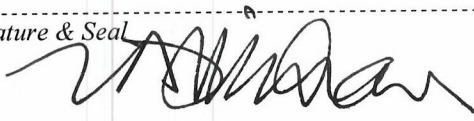
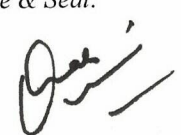
11. GOVERNING LAW OF THE AGREEMENT

This Agreement shall be governed by the Laws of Pakistan, as may be amended from time to time.

IN WITNESS WHEREOF, the Parties through their respective representatives who are duly authorized hereto have signed and agreed to this Agreement, all terms, and conditions of purchase order no **CPPA-G/2021/CIO/4396-98** and the Annexure: I thereto which form an integral part of this Agreement as of the date first written below.

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CPPA-G	M/S NEXTBRIDGE PVT LTD.
Name: ARSHAD JAVED MINHAS	Name: OMAR A. MIAN
Title: CHIEF INFORMATION OFFICER	Title: PRESIDENT
Signature & Seal:  Chief Information Officer Central Power Purchasing Agency (CPPA)	Signature & Seal: 
Date:	Date:

Witnesses	
Name: Muhammad Salwan Zahid	Name: MUHAMMAD ISHAR
Address: Shaheen Plaza, 73-West Fajale-hay Road, Blue Area, Islamabad	Address: 427/428-G4. M.A. JOHAR TOWN, LAHORE
CNIC: 32102-1022758-9	CNIC: 35202-2386911-9
Designation: Dy. Manager (Software)	Designation: CTO
Signature & Seal: 	Signature & Seal: 
Date:	Date:

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ANNEXURE: I

1. SCOPE OF WORK:

M/S NEXTBRIDGE PVT LTD. (the “Contractor”) will provide the **ISMO Website development (Documentation, Training, three years Domain registration & Hosting and Initially one year Support)** at the CPPA-G Islamabad. Project scope is to build ISMO online presence by developing web application for public and Registered Parties. This application would provide easy access to ISMO staff to share and post information. Through this application Registered Parties would be able to review their billing statements. Project scope is to build ISMO online presence by developing web application for public and Registered Parties. This application would provide easy access to ISMO staff to share and post information.

1.1 Purpose of Document:

This is an overview of the requirements for a new web-based solution for Independent Systems and Market Operators (ISMO). ISMO shall be unaffected by actions linked to power generation, ensuring that all generators are treated equally. This document not only describes the key features and functional requirements, but also the non- functional requirements. This document will lead the design and execution of the target system as well as the project's scope. Any requirement not mentioned in this document will be considered out of scope for this project.

1.2 Project Risk:

Following are some of the risks identified during project development:

- i. Delay in finalizing organization name which would affect designing logo and web application theme.
- ii. Delay in approving logo and theme design.
- iii. Delay in providing 3rd party APIs for integration in Public and Registered Parties portal

1.3 Development Phases:

ISMO online web application will be developed in 2 phases. In 1st phase all feature set related to CMS website will be completed. This includes Front-end application for public and admin panel for ISMO staff.

In Phase 2, Registered Parties portal will be developed along with 3rd party API integrations.

Complaint Management and Discussion Forum features will be included in phase 2 development.

2. CMS WEB FEATURES

2.1 ISMO Website Theme:

Front-End and Admin panel will have separate theme. For Front-End custom theme will be designed and webmaster will have option to select from 3 pre-defined theme. For admin panel, open-source Admin LTE responsive theme will be utilized.

Home page will be designed based on corporate colors closer to blue tone which is also used in CPPA-G website. Localized content should be displayed on website. Website should be designed considering minimalistic design approach. It should be clean and uncluttered designs.

In 1 theme, there should be 4 main menu categories on top menu and all other links will be displayed either on sub menu level or on footer.

ISMO website should be responsive hence its layout should also adjust on mobile or tablet screen accordingly.

On home page basic stats will be displayed and details should be on inner pages. Corporate information or About us info will in inner pages. Social media links will be on footer.

2.2 ISMO Dashboard:

ISMO staff/Content Creators would be able to login and view dashboard. On this dashboard following components/widgets will be available:

- i. Shortcuts / Quick links
- ii. Items requiring Actions.
- iii. Activity Log

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- iv. Recently updated information
- v. Content reaching expiration date.

2.3 Content Management:

Content Management module will have both options to either add content through HTML or user can simply add text & images and publish it on website. Only webmaster or Level 2 role can modify HTML. For HTML modification, level 2 role would be able to use general HTML tags like paragraph <p> and heading <h> tags.

Content Management module will provide following options:

- i. **Content States:** Following four states of content will be maintained:
 - Save as Draft
 - Published
 - Unpublished (For expired content)
 - Scheduled (For future scheduling)
- ii. **Content Modification:** Content Creator would be able to add, modify, remove, or move content using an easy-to-use editor.
- iii. **Content Preview:** Content Creator will be able to preview the content before publishing it. Content will be opened in new tab for preview. There will be no approval mechanism for content publishing as all content will be pre- approved.
- iv. **Content Expiry:** For time-sensitive content, user will be able to set an expiration date and time. User will also receive an expiration notice. This notice will also appear on the Admin's/Content creators dashboard.
- v. **Content Scheduling:** From admin panel content creator will have option to schedule CMS content, news, blogs etc. Content will be published on scheduled date and if expiry date provided, then content will be unpublished automatically.
- vi. **WYSIWYG Editor:** While offering flexibility for higher-end content contributors, the CMS will have a sophisticated WYSIWYG rich text editor for content additions and modifications that is easy and uncomplicated, giving content contributors a basic set of choices to alter/modify content.
- vii. **Banner:** Two options will be provided to upload banner. User can either upload new banner or add a link from media library.
- viii. **Spell Check:** Content Editor should have a spell checker that would help in eliminating any errors.
- ix. **User Friendly URL:** System will automatically generate user friendly URL's for newly created pages. URL routing will be provided along with meaningful URL's.

2.4 ISMO Library:

Documents related to regulations, announcements, reports etc. will be included in the ISMO Library. A CRUD will be provided in the admin portal along with a list to view the documents. Admin would be able to upload document or select already uploaded document. Admin/Content creator will have option to select checkbox to convert document into PDF. Only extensions docx (Document file) and ppt can be converted into PSD. Other extensions like csv or excel will not be converted. Visitors will be able to download the documents from front-end. If content was converted into PDF, then only pdf version will be downloaded not the original extension.

2.5 Graphics Management:

In this section, webmaster has option to add/edit images for different sections of front-end website. Webmaster will upload image along with title, description, meta tags, and keywords. Webmaster will be able to upload images with specific size and dimensions so that it does not affect front-end application layout.

2.6 Menu Management:

A dynamic menu structure will be provided that allows users to effortlessly add, update, move, and delete menu items across the site's structure. User will be able to add infinite number of menu levels. User will be able to handle the order of menus and sub-menu's manually. On front-end home page, top menu will have 4 main menu items which will have sub menus. Corporate / About Us info will be links on top of top menu.

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2.7 Broken Link:

System will display list of all broken links in admin panel. User will be able to view and rectify the link.

2.8 Search Statistics:

Site search statistics will be displayed in back portal. User can either export the results or filter and sort the results based on date and time criteria.

2.9 WEB Analytics:

Google analytics will be integrated with the website. Specific analytical data will be displayed in Admin portal dashboard. Type of data displayed in admin portal will be based on export permissions provided by Google Analytics.

2.10 Roles & Permissions:

Webmaster will have access to roles and permissions section where webmaster can create new roles and assign permissions. By default, Content Creator level 1 and level 2 roles will be provided along with super admin and registered parties role. Webmaster can modify permissions for these roles. Roles will be assigned to admin users. Admin users will have separate logins to access admin portal. Additionally, Department role will be created as well, which will be assigned to department accounts while adding departments.

Following roles will have these permissions by default:

- i. Webmaster: Has all permissions
- i. Level 1: Can only do Data Entry and content publishing and HTML part will be locked.
- ii. Level 2: Can have Level 1 access along with HTML changes.
- iii. Registered Parties: Access to Registered Parties portal
- iv. Department: Has access to department account in admin panel to manage complaints from Registered Parties.

Activity based logs will be maintained for all roles and webmaster will have access to logs.

NOTE: Webmaster can change role names and permissions as per need.

2.10 Admin User Accounts:

Webmaster will have option to create admin user accounts and assign roles to these accounts. Once added an email will be generated to the users email which would contain a link to add password to their account. After adding password, user can login to admin panel. Additional options will be provided to admin users to retrieve password or update password from admin panel. Webmaster can make user account Active or Inactive. In case account is inactivated, admin user will not be able to login into admin portal.

2.12 Automatic Sitemap:

A sitemap will be regularly updated. The sitemap and on-page breadcrumbs will be updated whenever the content creator adds, edits, or deletes any content.

2.13 PDF Convertor:

From admin panel, admin users will have option (checkbox) to convert non-PDF documents into PDF while uploading document. Only docx and ppts will be able to convert. It will be a background process. When Guest or Registered Parties try to download these converted documents, then automatically PDF version will be downloaded.

2.14 Design & Theme:

Landing page will be kept simple and sophisticated with images been localized. Look and feel of website will be clean, sleek, and minimalistic. All the pictures will be vetted by ISMO before adding onto website. In one of the templates, there will be four fixed main menus on the top of the screen. All these menus will have further sub-menus. Corporate / About us info links will be available on top of menu. Corporate / About us information will be on inner pages not on home page.

Three sets of pre-defined themes will be provided in back end. Theme change will be implemented on whole website. Content area layout will not be changed. Only appearance of website will be changed.

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3. COMPLAINT MANAGEMENT TOOL:

Complaint management tool will be available for Registered Parties to generate complaints and ISMO department to review and manage these complaints.

3.1 Complaint Management Tool for Registered Parties:

Registered parties will login to the RP portal using their own credentials. User information from the profile will be auto populated in the complaint form. RP will select department for the complaint from the drop-down menu. RP will fill out the complaint form and submit it. RP has option to upload any document for evidence as well.

RP would be able to perform following activities:

- i. Create new complaints
- ii. Select department
- iii. Enter details
- iv. Upload any document
- v. View List of all complaints generated by RP
- vi. View ticket status and details

From dashboard, RP would be able to view following stats:

- i. Total no. of complaints
- ii. Total no. of complaints resolved
- iii. Total no. of complaints unresolved

3.2 Manage Complaints:

Pre-defined department account will be provided to ISMO. When Registered Parties generates a complaint for any of these department, ticket will be automatically assigned and available for review in respective department accounts.

From department accounts, staff would be able to perform following activities:

- i. View list of complaints
- ii. View complaint details
- iii. Add comments
- iv. Download documents in complaint
- v. Upload any document
- vi. Mark ticket forwarded to external organizations. (ISMO will provide list of external organizations)
- vii. Update complaint status

On dashboard, ISMO departments would be able to view following stats:

- i. Total Tickets
- ii. Resolved Tickets
- iii. Open Tickets
- iv. Average Close time of ticket

3.3 Department Accounts:

Webmaster will have option in admin panel to create departments and add users in each department. These users will be assigned role i.e., Manager, PM or APM etc. When these users' login they will be able to see complaints for that department.

3.4 Email Notifications:

System will generate following email notifications:

- i. Email notification to RP when they generate new complaint.
- ii. Email notification to APM and CC PM for new complaint generated.
- iii. In case no action performed on ticket for 24hrs, email notification will be generated to PM and CC Manager.
- iv. In case no action performed on ticket for 48hrs, email notification will be generated to Manager.

3.5 Reports:

Departments will have option to generate report consisting of all tickets and their statuses.

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4. **REGISTERED PARTIES REGISTRATION:**

Registered Parties will be registered on ISMO internal systems. In phase 1, ISMO online website will have basic online RP registration FORM where registered parties would be able to register and provide company information. Along with providing information required for registration, registered parties can upload documents. On submissions, all these details entered in registration form will be downloaded automatically along with registration guide documents uploaded by ISMO.

From backend, admin would be able to view list and details of registered parties.

In Phase 2, APIs will be exposed to share registration details with ISMO internal systems. Once registered there, they can login to RP portal on successful authentication from ISMO internal systems. Authentication API for RP portal will be developed in phase 2.

5. **CONTACT US:**

A simple contact us form will be provided where user will enter his personal details along with query. In admin panel, every submitted query will be listed under contact us section. ISMO staff would be able to view list and details of each query.

6. **E-NOTIFICATIONS:**

Registered Parties will have the option to receive important notices through email. They will have the ability to opt in/opt out to receive e-notifications. To receive e-notifications, a valid and verified email address is mandatory.

7. **E-NEWSLETTER:**

E-Newsletter subscription service will be provided so that users can receive newsletter through email. Guests can subscribe for newsletter by provide name and email. From admin panel, ISMO staff would be able to see list of users subscribe for newsletter and send them email content.

8. **EMBEDDED AUDIO/VIDEO/MEDIA:**

Easy embedding of audio, video and media with associated embed code will be provided.

9. **FAQ TOOL:**

A FAQ section will be provided on front-end website. ISMO would be able to add FAQ from admin panel. ISMO would be able to publish and unpublish FAQs.

10. **GIS MAPPING (OPTIONAL):**

Initially this feature will not be implemented. This could be required in support period. A map will be displayed with locations marked on it.

11. **IMAGE MANAGEMENT:**

Image management tools for the addition of images to on-site content.

- i. **Editing:** Image editing abilities on uploaded images, including the ability to change opacity, resize images dynamically based on width and height, flip images, rotate images, and crop images.
- ii. **Multiple Upload:** Capacity to upload multiple images at one time and associate images with specific pages; the maximum file size should be no less than Two (2) megabytes.

12. **JOB POSTS:**

ISMO HR will have option to add, edit and delete job ads by providing job details. Job title, Job description, Experience Required, Qualification and How to Apply will be posted on Front end website. Images and documents can also be uploaded while posting the job. HR would be able to enable or disable standard FORM to apply for the job online. If enabled, candidate will have to provide basic details along with CV to apply for job. HR would be able to view list of all candidates that have applied for job for each job ad. There will be ability to extract list of candidates and details in CSV.

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13. LANGUAGE TRANSLATION:

ISMO website would be multi-lingual website. Primary language would be English. Secondary language would be Urdu. Google translator will be used to convert language to Urdu.

14. ONE-CLICK SOCIAL MEDIA:

Social Media accounts Facebook, Twitter and LinkedIn will be integrated with the website. Icons will be provided in the footer of website. Front End will only display social icons with URL's linked to them. One click posting is dependent on third party API's. Content Creators would be able to share content on social media after publishing it. Content Creators will have to share content separately for each social media directly from admin panel. Social Feed will not be displayed on front-end.

15. PHOTO SLIDESHOWS:

On Front-end website there will be section for Photo Slideshow. ISMO staff will have option in admin panel to upload photos or choose from media library. Also they have to alter the order, speed, transition type and duration of on-site slideshow from back end.

16. SOCIAL MEDIA INTEGRATION:

Front End will only display social icons (Facebook, Twitter and LinkedIn) with URL's linked to them. Social Feed will not be displayed.

17. NEWS POSTING:

A dedicated news section will be provided where website users can view latest news related to ISMO and ISMO activities. ISMO will have options to publish/unpublish news from admin panel. They can also save content as draft. An editor will be provided that would allow content creators to add and modify text, add and resize images and spell check. When news or any event is posted it will relate to RSS feed.

18. ACESSEBILITY ADD-ONS:

Following accessibility add-ons must be provided on the website:

- i. Change Font size.
- ii. Audible content.

19. ADVANCED SITE SEARCH:

Search will be based on metadata. Internal search will have following functionalities:

- i. **Sort:** Users should be able to sort search results by date, content, title, or relevance.
- ii. **Filter:** Users should be able to filter by type of content and easily apply advanced search techniques such as Boolean, if desired.
- iii. **Search Optimization:** User should be able to tune the search results by using synonyms for common words or terms and promote pages using keywords.
- iv. **Document Search:** Search functionality would search title and description provided before uploading any document. Based on it search results will be displayed to public.
- v. **Internal Search:** Search will be based on ISMO website and no external search from google, or other sites will be incorporated.

20. 3rd PARTY API INTEGRATION:

In ISMO online website few APIs will be integrated from 3rd party systems. Stats and data displayed on landing page and inner pages will be displayed using these API integrations.

Following are some stats that will be displayed on landing page:

- i. Load Forecast
- ii. Load Data
- iii. Marginal Price

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Details along with other stats will be displayed on inner pages. Similarly, stats and details displayed for each Registered Parties will be displayed using 3rd party API. API integration will be part of phase 2 development after ISMO internal software are developed and APIs are available. Additionally, few APIs like RP registration API will be exposed to other systems. These APIs will be developed and discussed in later stages of development.

21. RSS FEEDS:

Guest user should be able to subscribe for RSS feed. Through RSS feed user would be up to date on important events, news, and announcements from the website.

22. MEGA MENUS:

The solution should provide capability for fully customizable mega menus. System would automatically create menu based on content published. ISMO will have option to add, remove and change order of navigation.

23. IFRAME FUNCTIONALITY:

The solution should have iframe functionality to seamlessly embed other documents within any HTML page. Examples of embedded content include videos, third party applications, Slide Share documents, etc.

24. DOCUMENT ARCHIVE:

A document archiving feature would be available where ISMO can archive documents into different categories. If these documents are published, then they will be displayed on front-end and will appear on search results. Guest users can download PDF documents from archived section by providing Name and Email.

25. REGISTERED PARTIES PORTAL:

In Phase 2 Registered Parties portal will be developed. Only Registered Parties will be able to access this portal to review Preliminary Settlement statement, balancing mechanism of energy results, extraordinary settlement statement and final settlement statement.

25.1 Authentication: Registered Party user will get authenticated from ISMO internal system. RP will be login successfully into RP portal when successfully authorized from ISMO internal system.

25.2 Reports: Registered Parties would be able to review reports. All data will be fetched from ISMO internal system and displayed on RP Portal.

26. CHATBOT:

ChatBot will be developed for guests on ISMO online website. On bottom right corner of ISMO website there will be an icon for ChatBot. Clicking that icon will open small window for ChatBot. User will be able to ask query and ChatBot will answer accordingly to available knowledgebase. Before user can start chat, they will have to provide Name, Email, Company (optional) and phone number. User will also have option to enable whether to receive chat transcript in their email. If Bot will be unable to answer the query, then user will have option to leave message for ISMO staff. A description box will appear that would allow user to add description. Query will be added to contact us list in admin panel. From admin panel ISMO staff can view details about the query. In admin panel, ISMO staff will have option to add question and answers which would become knowledge base for ChatBot. For all ChatBot sessions, system would capture chat and send it to ISMO specified email address.

27. DISCUSSION FORUMS:

Discussion forums will be available for Registered Parties and ISMO staff. ISMO staff will be able to create new forum threads. Forums can be published, unpublished or closed. Published forums will be displayed on RP portal. ISMO staff will be able to add, hide or remove comments. Registered Parties would be able to view list of forums along with their details. They would also have option to comment on these forums. In phase 1 layouts for front-end discussion forum will be designed but actual development will be in phase 2 after completion of other modules.

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28. NON-FUNCTIONAL REQUIREMENTS:

28.1 WEB Hosting:

Web hosting service will be provided for this project. Server will be hosted on 3rd party server from Hetzner. Nextbridge will configure web hosting server with following specs:

- i. 4 GB Memory
- ii. 2 Core Processor
- iii. 80 GB SSD Disk
- iv. 4 TB Transfer

NOTE:

- All terms and conditions from Hetzner will be applicable and subject to change (<https://www.hetzner.com/legal/terms-and-conditions>)

28.2 DDoS Mitigation:

Web hosting companies automatically provide protection from DDoS attack. Similarly, for ISMO website, Hetzner will provide automatic safeguard to defend against most common, frequently occurring network and transport layer DDoS attacks that target web site or applications.

28.3 Backup & Disaster Recovery:

- i. We will be taking data and database backups on daily basis which will be saved on an additional volume.
- ii. We will also enable automated daily backups of the server which will have everything in it. There will be last 5 backups available at any given time
- iii. In case of a disaster, we will
 - Restore the entire instance from one of the automated backups taken in clause 2 - this usually takes a few minutes.
 - Create a new instance and then refresh the data & database from the backups we have taken in clause 1 - this usually takes anywhere between 1 hour to 6 hours (depending on the size of the backups and the day of the week*)

NOTE:

- In case of any issue which requires recovery from these backups; timely intimation to Nextbridge is required.
- Usually, we have an availability window of about 14 hours on weekdays BUT during weekends, it may take 2-4 hours for an engineer to be available for any emergency maintenance.

28.4 Browser Support:

New ISMO website will be responsive, and it will support following browsers:

- Apple Safari
- Google Chrome
- Microsoft Internet Explorer and Edge
- Mozilla Firefox

28.5 Page Load Time:

The solution will ensure that page's load in 1.5 seconds or less on average.

28.6 System UP Time Guarantee:

99.9% system uptime will be provided.

28.7 Development, Test & Live Environment:

A contractor will establish development and test environment and will hand over to CPPA-G IT team so that the CPPA-G IT team may use them for future enhancements and support. All project details will be handed over to the CPPA-G IT personnel after the project is completed. During development Nextbridge will provide testing/demo environment using internal Nextbridge servers. Once application goes live, Nextbridge will provide staging environment on Hetzner servers.

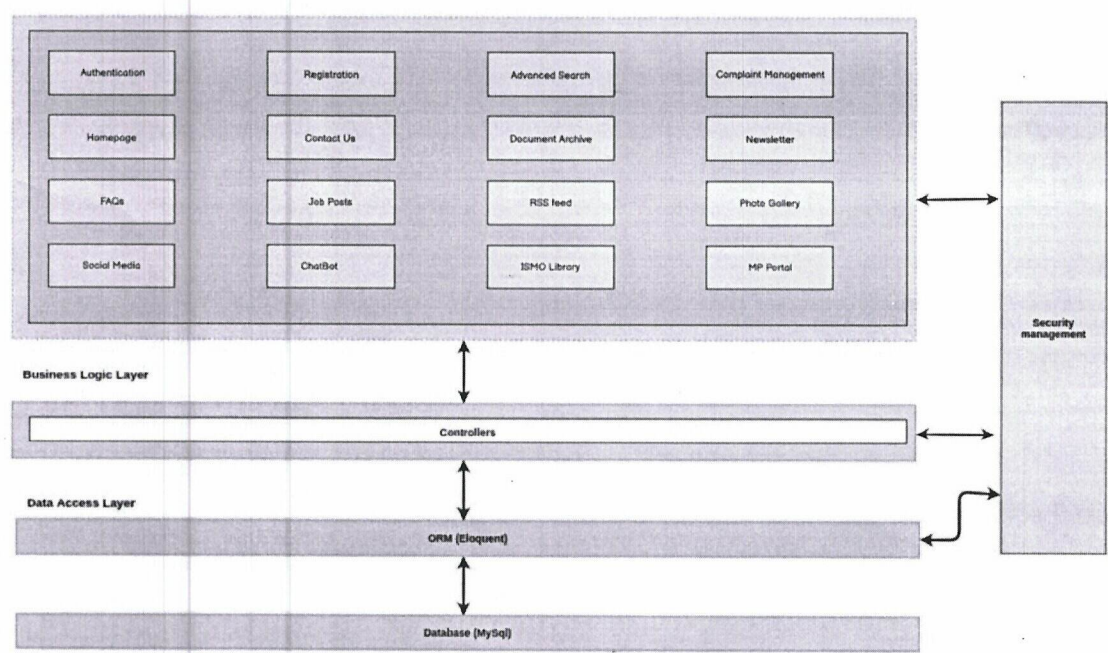
28.8 SSL Certificate:

SSL certificate will be installed on server to encrypt data in site communications.

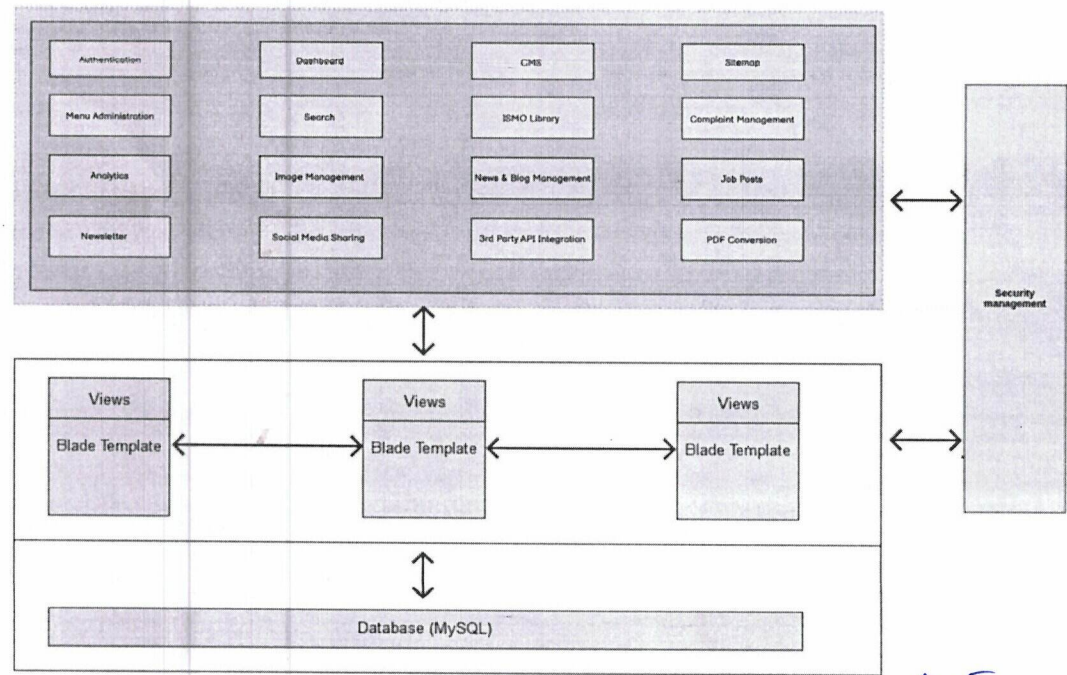
29. WEBSITE ARCHITECTURE:

For ISMO online website will layered and three tier architecture.

➤ Front-end ReactJS Architecture Diagram



➤ Back-end Architecture Diagram



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30. **TECHNOLOGY STACK:**

Following is technology stack for ISMO online website:

- ReactJS
- PHP, Laravel
- MySQL
- JavaScript
- HTML/CSS

i. **Front-End Website Structure:**

- Site will be a Single Page Application which will be built using ReactJS components.
- Layout will be completely responsive. Consistent display will be provided across all resolutions to cope different devices like mobile phones, tablets and laptops.

iii. **Back-End Website Structure:**

Application backend will be built using a widely used PHP Framework Laravel. Laravel is known for its MVC architectural pattern, Modular Packaging System, dedicated dependency manager and for the different ways available in it to access and communicate with relational databases.

iv. **Administration Panel:**

Admin panel is the heart of the back-end access to the Web Portal. Available only through secure login, this is a place where admins would manage the front-end website content. Through this application, the admin would be able to authorize and deploy new content uploaded.

v. **Database:**

All information submitted and received goes through a database housed on the same server that serves the website. By employing the database (either SQL Server or MySQL) the portal is constantly dynamic.

vi. **Application Architecture:**

Application will be built using layered architecture approach. ReactJS will be used as a client-side framework, which will make site highly interactive and independent from business logic. All business logic will be present in the rest API. And all data will be stored in a secure MySQL DB. Separation of concerns among these logical layers and components provide increased flexibility, maintainability, and scalability.

31. **TECHNICAL REPORTS:**

The Contractor will provide all maintenance reports performed, including the routine preventive maintenance practices and fault reports, as required by CPPA-G and shared between CPPA-G’s technical team and the Contractor’s service teams as necessary. Reports required by CPPA-G’s management regarding the status of the system and any issues relating thereto will be provided by the Contractor as and when required.

32. **POC AND ESCALATION MATRIX:**

The Contractor shall provide the information required, including the Escalation Matrix, as per the tables below. In case there is any change during the tenure of this Agreement, the Contractor shall provide the details of the same to CPPA-G without delay.

Support Contact			
Support	Name	Contact No	Email Address
First Level Support	Tehseen Ahmad	03103336365	tehseen.ahmad@nxb.com.pk

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Escalation Matrix				
Escalation Level		Name	Contact No	Email Address

First Level Escalation	Tehseen Ahmad	03103336365	tehseen.ahmad@nxb.com.pk
Second Level Escalation	Omer Farooq	03224625612	omer.farooq@nxb.com.pk

Helpdesk		
UAN		92-42-35315043
Tel		03103336365
Email		tehseen.ahmad@nxb.com.pk
Address		428-G M.A. Johar Town, Lahore

33. SUPPORT SEVERITY LEVEL:

Depending on the effect of a problem, following are the defined four severity levels:

- i. **Severity Level 1 – Minimal Business Impact:** Customer submits a software information request, software enhancement or clarification which has no operational impact. The implementation or use of the software by the client is continuing and there is no negative effect on the productivity. **Nextbridge (Pvt.) Ltd.** will provide an initial response regarding request within two (2) business days.
- ii. **Severity Level 2 – Some Business Impact:** Important features of the software are not working properly and there are no acceptable alternative solutions. While other areas of the software are not impacted, the reported has created significant, negative impact on client’s productivity or service level. **Nextbridge (Pvt.) Ltd** will commence work on resolve problem within 24 hours of notification and will engage staff during business hours until an acceptable resolution is achieved.
- iii. **Severity Level 3 – Significant Business Impact:** Important features of the software are not working properly and there is no acceptable alternative solution. While other areas of the software are not impacted, the reported problem has created a significant, negative impact on client’s productivity or service level. **Nextbridge (Pvt.) Ltd** will commence work on resolving the problem within 12 hours of notification and will engage staff during business hours until an acceptable resolution is achieved.
- iv. **Severity Level 4 – Critical Business Impact:** This impact of the reported deficiency is such that the customer is unable to either use the software or reasonably continue work using the software. **Nextbridge (Pvt.) Ltd** will commence work on resolving the problem within 4 hours of notification and will engage staff during business hours until an acceptable resolution is achieved.

34. RESOLUTION TIME(S):

CPPA-G will designate the CPPA-G Point of Contact (the “POC”) and will determine the Priorities to be notified by the POC to the Contractor. The POC will requisition replies / updates from the Contractor on any outstanding Problems as and when required by CPPA-G. The resolution time for each severity is as below:

No.	Problem Severity	Resolution Time
1	Level 1	48 Hours
2	Level 2	24 Hours
3	Level 3	12 Hours
4	Level 4	4 Hours

35. IMPLEMENTATION/ONSITE SUPPORT SERVICE FEATURES:

- i. The Contractor shall provide unlimited corrective maintenance (depending on the severity level as per clause 33 and 34 of Annexure I) twenty-four (24) hours a day, seven (7) days a week, three hundred and sixty-five (365) days a year, including on public holidays.
- ii. The response times for the resolution of Problems shall be as provided in clause 34 of this Annexure: I upon receipt of notice thereof by the Support Contact of the Contractor as per clause 32 of this Annexure: I.
- iii. The Contractor shall conduct Preventive Maintenance on a quarterly basis and submit a report on the same to CPPA-G with recommendations (if required).
- iv. The Contractor shall ensure that all Defective System(s) will be rendered fully functional and operational in the Resolution Time(s) provided hereinabove at clause 34 of this Annexure: I as per their respective Problem Severity Level. Noncompliance by the Contractor of this obligation shall lead to the imposition of liquidated damages by CPPA-G upon the Contractor as per this Agreement and clause 1.2 of this Annexure: IV.
- v. In event of non-performance by the Contractor of any of its obligations under this Agreement or the Annexures thereto, CPPA-G shall be entitled to deduct one half of one percent (0.5%) of the total value of the total value (inclusive of GST) of the Project daily from the Performance Security. CPPA-G further reserves the right to avail any or all remedies or actions available to it under the law
- vi. Any liquidated damages imposed under this Agreement shall be capped at an amount of five percent (5%) of the total value (inclusive of GST) of the Project per year. This clause shall not prevent CPPA-G from imposing liquidated damages determined to be liable in the preceding agreement year in the subsequent agreement year.
- vii. CPPA-G shall ensure that Personal designated by the Contractor is provided timely access to the resources required. The time keeping conducted by the CPPA-G Coordinator shall be consider as final.
- viii. Following the occurrence of any Problem the Contractor shall submit a detailed Root Cause Analysis to CPPA-G, along with its recommendations on how to prevent the reoccurrence of such Problem.
- ix. Following the occurrence and subsequent resolution of a Problem, the Contractor shall log and conduct all preventive measure(s) to be taken to prevent the reoccurrence of such Problem.
- x. If CPPA-G is dissatisfied with the support services of the technician(s), engineer(s), and / or other staff assigned certain responsibilities under this Agreement by the Contractor, CPPA-G has the right to submit a written complaint to the Contractor, detailing the cause of such dissatisfaction with the Contractor's personnel's performance and requiring the replacement of such personnel.
- xi. This Agreement shall also apply to Implementation and Documentation, Training, Three years Domain registration & Hosting and Initially one year Support.
- xii. Provision of
 - Regular automation
 - Troubleshooting fixes & software release upgrades.
 - On call & onsite support of local vendor.
 - On-site (Principal backed) support for patches, upgrades, and configurations etc.
- xiii. Track and Record of all the issues and case submitted by the team for CPPA.

36. APPROVAL OF CHANGES:

When possible, changes to the client's site will be posted under a test environment and a URL will be sent to client to review these changes. If necessary, changes will be made directly to the live site and client will be asked to review the changes immediately. It is the responsibility of the client to review changes in a timely manner and ensure that the changes are correct.

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ANNEXURE: II

1. PROJECT PLAN:

The project plan for the required services is given below:

1.1 Phase-1 Development:

In Phase 1 development following feature set will be developed:

- i. Logo and responsive theme
- ii. Dynamic Navigations
- iii. Static Pages, Contact Us
- iv. News, Blogs, and other content
- v. Advance Search
- vi. Multi-Lingual
- vii. ISMO Library
- viii. Site Map
- ix. FAQs and Career
- x. Photo and Video Gallery
- xi. Social Media Links
- xii. E-Newsletter and RSS
- xiii. RBAC Control
- xiv. Content Management
- xv. ISMO Dashboard
- xvi. Google Analytics
- xvii. PDF Conversion
- xviii. Document Archiving
- xix. ChatBot
- xx. Web Form for Registration of Parties (i.e. Registered Parties)

1.2 Phase-2 Development:

In phase 2 development all 3rd API party integrations along with Registered Parties portal will be developed. Phase 2 of ISMO development will start once CPPA-G internal system is ready so that ISMO website can be integrated through APIs. Once APIs are provided and ISMO gives approval the phase 2 development will be initiated.

Following features will be part of phase 2 development:

- i. Registered Parties Portal
- ii. Integration with 3rd Party API for public and Registered Parties portal
- iii. Complaint Management Tool
- iv. Discussion Forum

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ANNEXURE-III

1. EFFECTIVE DATE OF COMMENCEMENT:

Effective date of Commencement of Service under this Agreement is 01-02-2022 (1st February 2022).

2. TIME SCHEDULE OF SERVICES:

The Time schedule of Services for milestones / deliverable under is agreement is given below:

PHASE -I	
1 st February, 2022	Sprint 1 (DB Schema & Application setup, Authentication, News Management, ISMO Library, Job Posts)
14 th February, 2022	Sprint 2 (CMS, Image Management, FAQs, Theme Change Ability)
28 th February, 2022	Sprint 3 (Menu Management, Breadcrumb, Newsletter, Photo/Video Gallery, FAQ page (API), DB Schema & Project Structure, Integrate UI)
14 th March, 2022	Sprint 4 (Automatic Sitemaps, Front end pages, Newsletter, ISMO Library, Job Posts, CMS Pages, Contact us and sitemaps, FAQ front end)
28 th March, 2022	Sprint 5 (RSS Feed (API), Advance Search, Dashboard, ISMO Library front end, Newsletter, job posts, RSS feeds (front end))
11 th April, 2022	Sprint 6 (Broken links, social media icons, PDF conversion, Advance Search, Multi-Lingual, Social Media links, Chatbot)
25 th April, 2022	Sprint 7 (Graphic Management, Web Analysis, Adjustable Front End, Dashboard/ Home Page)
16 th May 2022 – 29 th May, 2022	Sprint 8 (Testing, Deployment)

PHASE -II	
1 st August, 2022	Sprint 9 (Authentication API and front end, Discussion forum API and front end, testing)
15 th August, 2022	Sprint 10 (Complaint Management system API, front end, testing, M.P portal)
29 th August, 2022	Sprint 11 (Third party API integration)
12 th September 2022 – 31 st September, 2022	Sprint 12 (Deployment)

3. EXTENSION OF TIME:

Extension of Time for completion of Services and the terms & conditions thereof shall be mutually agreed between the Client and the Contractor as and when required.

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ANNEXURE-IV

1. PAYMENT FORM:

Implementation Cost under this agreement will be made after the acceptance of deliverables as per the following Schedule:

S No.	PROJECT PHASE	Deliverable / Phase Output	Payment Terms
1.	Mobilization Cost	Project Scope Project Plan	15%
2.	Completion of Phase 1 (After Testing & Acceptance & GO-LIVE)	Sprint 1 – Sprint 8	35%
3.	Completion of Phase 2 (After Testing)	Sprint 9 to Sprint 12	30%
4.	Training of CPPA-G IT team (after ISMO Website Published) Documentation Handing Over to CPPA-G	Technical Training to CPPA-G IT Team. All project related documentation including user manual.	10%
4.	Performance Evaluation Cost after one month of Go-LIVE		10%

Notes:

- i) All rates are inclusive of the cost(s) of inspection, delivery, installation charges, services, transportation, all applicable taxes, import duties if any and other levies.
- ii) All prices are in PKR.
- iii) The annual price in PKR shall remain firm and final and shall not subject to any escalation during three contractual years.

Special Conditions

- i) M/S NEXTBRIDGE PVT LTD. shall be responsible for completing the task(s) prescribed in the work order in accordance with the methodology and in the time, period as specified in the Agreement. Should M/S M/S NEXTBRIDGE PVT LTD. fail to do so it shall be liable for liquidated damages as per the Agreement.
- ii) M/S NEXTBRIDGE PVT LTD shall be responsible for all expenses for service and delivery and for any other software, license utility software, third party tool etc. required for the delivery of service(s) under the Agreement.

1.1 TERMS OF PAYMENT:

Payment of the Milestones/Deliverables/Services/Implementation will be made directly by consignees within thirty days, from the date of receipt of invoice, on production of following documents:

- i. All prices are in PKR.
- ii. The price in PKR shall remain firm and final and shall not subject to any escalation during agreement validity period.
- iii. All rates are inclusive of the cost(s) of inspection, delivery, implementation charges, services, transportation, all applicable taxes, import duties if any and other levies.
- iv. Invoice in triplicate having NTN, GST No. printed of both Supplier & Consignee.
- v. Deliverable Acceptance certificate issued by CPPA-G.

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- vi. Sales Tax Invoice (if not exempted) or attach exemption certificate.
- vii. Professional Tax Paid Certificate by the firm.
- viii. For release of GST Payment Supplier should provide copy of submitted Annex-C from FBR Portal

1.2 LIQUIDATED DAMAGES:

In case M/s Nextbridge Pvt Ltd. fails to deliver the services thereof within the specified delivery period, CPPA-G will be entitled to.

- i. Failure to deliver any or all the components of Website within the stipulated time specified, then CPPA-G may deduct from the invoice a sum of money equal to the 2% per week of the specific project phase between the scheduled delivery date and the actual delivery date(s) up to maximum of 10% of the total project phase value.
- ii. Failure to deliver accurate website within the stipulated time period will invoke a penalty as specified in above sub-section (i); however, in case of failure to delivery, Performance Security amount may be forfeited, and the company may be blacklisted and may not be allowed to participate in future tenders as well.

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